Fiscal Year:	FY 2016	**Task Last Updated:**	FY 05/03/2016
PI Name:	Chahine, Georges D.Sc.		
Project Title:	Microgravity Fluid Separation: Dynaswirl® Bubble Vortex Separator		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Fluid physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	20794	**Congressional District:**	7
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2008 Fluid Physics NNH08ZTT002N
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No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:	NOTE: Extended to 2/24/2016, per NSSC information (Ed., 10/7/15) NOTE: to be extended, per B. Motil/GRC (Ed., 6/26/15)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Wu, Xiongjun (Dynaflow, Inc.) Loraine, Greg Ph.D. (Dynaflow, Inc.)		
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Performance Goal Text:			

Task Description:	The limits on the amount of liquids and gases carried to space make it imperative to recycle and reuse. On Earth, bubbles are relatively easily separable by buoyancy. In microgravity, other external forces such as a centrifugal force must be utilized to separate bubbles from liquids. Two categories of centrifugal force separators exist: one uses active rotation by mechanically spinning the tank. This is very efficient but requires shaft, bearings, and motor. In the second, the tank is fixed and the rotation is induced by eccentric injection of the mixture (Free Vortex Separator or FVS). This FVS has been proven very efficient for mid-range void fraction (50-80% gas) with an efficiency approaching 100%. However, this FVS is not as efficient for lower void fraction, and there is a need of a separator for low void fractions. We have developed a phase separator at DYNAFLOW based on a DYNASWIRL® swirl flow nozzle. It consists of two concentric cylinders, the fluid is introduced in the space between the two cylinders, and the swirling flow inside the inner cylinder is generated with the help of tangential injection slots. With this configuration one can make the vortex circulation intense enough to generate cavitation with a very stable vaporous/gaseous core even at low flow rates. Operation with the high swirl near or slightly below cavitation inception in the vortex core guarantees removal of air even from mixtures with very low void fractions. This is achieved by forcing the bubbles to expand and collect in the vortex core. These are then removed through two central exit orifices at each end of the cylinder. This configuration was tested successfully during a first reduced-gravity flight and we are planning additional testing this month. We are further developing and experimentally testing this configuration to characterize its gas-liquid separation performance and to improve its efficiency. We are also conducting numerical simulations to help optimize the design and ultimately build a compact system for space validation on the International Space Station (ISS) using the existing Fluid Integrated Rack facility. NOTE: Continued by "Two-Phase Flow Separation and Characterization for Microgravity Applications" grant NNX16AB85A with same Principal Investigator Georges Chahine, D.Sc. (Ed., 5/4/2016)
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	As a passive separator device, the DYNASWIRL® phase separator has the advantage of simplicity and dependability. It requires no moving mechanical parts and low power. Compared to other passive separator devices, which usually are not very efficient at low void fraction, the DYNASWIRL® is able to force low void fraction gas out of the liquid into the central core of the vortex by combining the effects of swirl, cavitation, and rectified gas diffusion. In addition to being an effective phase separator for space applications, it will also find wide applications in Earth applications where fast separation of low void fraction gas is of concern.
Task Progress:	During this reporting period, we have focused our efforts on testing and improving the performance of the small scale DYNASWIRL® phase separator, which was then integrated in the NASA Glenn Research Center (GRC) breadboard test rig. Systematic ground based experiments and numerical simulations were performed before the test section was delivered to NASA. The new phase separator was first tested in a NASA reduced gravity flight campaign on June 23-25, 2015. Following these tests, the gas extraction scheme was further improved and the phase separator was tested again in a second NASA reduced gravity flight campaign on December 14-16, 2015. We actively participated in both reduced gravity flight campaigns spearheaded by NASA GRC. Additionally we have continued our efforts to develop the mixture impedance based device to measure mixture high void fractions. A. Improvement of Gas Extraction Scheme Based on the laboratory and reduced gravity test results, we have further fine-tuned the liquid and gas extraction schemes. Main improvement included: (a) A single central orifice configuration was selected over off-centered multi-orifices following systematic tests on the liquid exit orifice layout. (b) An improved process to make gas extraction tubes with larger pore sizes was applied and allowed higher gas flow rates through the membrane while keeping the threshold pressure for water breakthrough high enough to minimize the chance of liquid leakage through the membrane. c) The size of the disk/vortex blocker size was optimized through systematic tests, and the effects of this size on the performance were determined. d) A vortex killer was introduced at the liquid exit to prevent the swirl flow from continuing downstream of the separator. B. Reduced Gravity Flight Test on June 23-24 2015 The small scale DYNASWIRL® phase separator was first tested in reduced gravity flights on June 23-24, 2015. Personnel from DYNAFLOW participated in the flight tests together with the NASA team. Different operation conditions were tested in over 80 flight parabolas. The tests showed that the gaseous core formed under all flow conditions tested. However, under extreme gas flow conditions, the gas extraction was not fast enough to evacuate the gas collected on the separator axis, and some gas could escape into the liquid line and contaminate the ABS measurements. The test data obtained during the reduced flights provided useful information, which helped further improve the gas extraction scheme as tested later in the second reduced gravity flight campaign in December as reported in Section E. C. Numerical Simulations to Help Optimization of the Phase Separator Operation 3D Numerical simulations were conducted to simulate the two=phase flow in the separator. 3DYNAFS-VIS© two-phase flow Navier-Stokes solver modeled the overall flow field of the bubbly mixture, while 3DYNAFS-DSM©, a bubble tracking and bubble dynamics motion solver modeled the bubble capture and behavior. Both components are modules of our 3DYNAFS © CFD suite and are coupled at each time step. The results from the numerical simulation were used to investigate the effects of gravity, to optimize the separator components geometry and the placement of several components such as the vortex blocker, the vortex killer, and the gas extraction tube. The numerical simulation results also provided invaluable guidance for determining the test matrix for the laboratory and flight experiments. D. Instrumentation Development for Void Fraction Measurement We continued the development of an electric impedance based void fraction measurement probe in an effort to build an

	instrument that can reliably measure the local void fraction of the two-phase flow mixture at both the inlet and outlet of the separator when the inlet void fraction is high. To validate the method, a void fraction loop was built and has fast acting pneumatic valves to capture a section of the two-phase flow to enable a direct measurement of the amount of air in the mixture. These measurements correlated well with the impedance probe measurements. Extensive experiments were conducted at various flow conditions and enabled generation of calibration curves of void fraction versus impedance probe readings at different flow rates. E. Reduced Gravity Flight Tests on December 14-16, 2015 Following the previous reduced gravity flight campaign in June, efforts were made to improve the extraction tube gas removal rates. After extensive testing, the membrane pore sizes were increased, the length of the gas extraction tube was increased, and the shape and size of the vortex killer and vortex blocker were modified. The updated DYNASWIRL® phase separator was again delivered to NASA GRC in November 2015 to be integrated with the NASA breadboard rig and NASA conducted independent ground tests. The rig was then taken for reduced gravity flight tests, which were performed in Houston on December 14-16, 2015. Personnel from DYNAFLOW participated directly in the flight campaign and conducted phase separation tests together with the NASA team. The phase separator performed well and lesson learned included items such as improvement in the high-speed photography of the phenomena and better optical access to the swirl chamber. This provided valuable information on how to further improve the DYNASWIRL® phase separator, which led to a new design of the test section, which is undertaken under the follow-on contract. NOTE: Continued by "Two-Phase Flow Separation and Characterization for Microgravity Applications" grant NNX16AB85A with same Principal Investigator Georges Chahine, D.Sc. See that project for further reporting. (Ed., 5/4/2016)
Bibliography Type:	Description: (Last Updated: 09/04/2025)
Articles in Peer-reviewed Journals	Hsiao CT, Ma J, Chahine GL. "Numerical study of gravity effects on phase separation in a swirl chamber. " Chemical Engineering Science. 2017 Jun 29;165:177-85. https://doi.org/10.1016/j.ces.2017.03.005 , Jun-2017
Articles in Peer-reviewed Journals	Wu X, Loraine G, Hsiao C-T, Chahine GL "Development of a passive phase separator for space and earth applications." Separation and Purification Technology. 2017 Dec;189:229-37. https://doi.org/10.1016/j.seppur.2017.08.017 , Dec-2017
Papers from Meeting Proceedings	Wu X, Loraine G, Hsiao C-T, Chahine GL. "Development of an efficient phase separator for space and ground applications." To be presented at ASME 2016 Fluids Engineering Division Summer Meeting FEDSM2016, Washington, DC, July 10 – 14, 2016. Proceedings of the ASME 2016 Fluids Engineering Division Summer Meeting FEDSM2016, Washington, DC, July 10 – 14, 2016. In press as of May 2016. , May-2016
Papers from Meeting Proceedings	Hsiao C-T, Ma J, Chahine GL. "Numerical study of gravity effects on phase separation in a swirl chamber." To be presented at ASME 2016 Fluids Engineering Division Summer Meeting FEDSM2016, Washington, DC, July 10 – 14, 2016. Proceedings of the ASME 2016 Fluids Engineering Division Summer Meeting FEDSM2016, Washington, DC, July 10 – 14, 2016. FEDSM2016-7845. In press as of May 2016. , May-2016